

Question Paper

Exam Date & Time: 29-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026

SUBJECT: ICT 3121/IT_3121 INFORMATION SECURITY

(SPL: INFORMATION TECHNOLOGY)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) The AES S-box is generated by taking the multiplicative inverse of each byte in $GF(2^8)$ (with the irreducible polynomial $x^8+x^4+x^3+x+1$) and then applying an affine transformation. (5)
Given the byte $b=0x57$,
a) Compute its multiplicative inverse in $GF(2^8)$.
b) Apply the AES affine transformation to obtain the SubBytes output.
c) Show all intermediate steps.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 1B) A company's internal emails are intercepted by an attacker who monitors message headers and communication frequency without altering any content. Identify the type of security attack represented in this scenario and explain which security principle (confidentiality, integrity, or availability) compromises and how. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) An attacker performs a chosen-plaintext attack on an Affine cipher and obtains: (2)
 $P=A \rightarrow C=I$
 $P=B \rightarrow C=N$
Later the attacker intercepts the ciphertext: SWFRCPQAXC
Determine the Affine cipher key (a,b) and decrypt the ciphertext.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) A sender wants to send a confidential message using the ElGamal cryptosystem. The public parameters are a prime number 67, e_1 is 2, the public key is 228, and the private key is 153. (5)
a) The plaintext message to send is 123. The sender chooses a random number 51. Compute the ciphertext.
b) Recover plaintext from (354,323)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2B) Two companies are evaluating cryptosystems to secure their communications. Company A operates in a high-frequency trading environment where computational speed is critical, as any delay in encryption or decryption could lead to financial losses. Company B, on the other hand, handles sensitive legal documents where the accuracy and uniqueness of decrypted messages are paramount, and even a small ambiguity could have serious legal or operational consequences. (3)
a) Compare RSA and Rabin in terms of their decryption process and implications for these two requirements.

b) Recommend which cryptosystem is better suited for each company and justify your reasoning.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2C) A healthcare company wants to implement a secure application to protect the integrity of healthcare data. State three essential criteria that a cryptographic hash function must satisfy to prevent an attack from an eavesdropper (Eve). (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3A) Compare and contrast the design of SHA-512 and Whirlpool hash functions in terms of their internal structure (rounds, word size, and block processing), design approach (custom compression function vs. block cipher-based), and their mechanisms for ensuring collision resistance and message integrity. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3B) Dr. Priya uses the El Gamal signature scheme to sign important lab reports. She selects the system parameters:
• Prime $p=19$
• Primitive root $e_1=10$
Priya's private key is $d=16$; she publishes the corresponding public key. Compute her public key. A collaborator, Arjun, wishes to sign the numeric report $M=14$. He (or Priya, if signing herself) uses a random ephemeral value $r=5$. Compute the El Gamal signature (S_1, S_2) for the message $M=14$. Show the verification calculation and confirm whether the signature is valid. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3C) For the scenario illustrated below, identify the type of forgery that has occurred in the digital signature system and justify your answer based on the attacker's capability and control over the forged message. (2)
- a) Before obtaining Diya's public key, an attacker pre-selects a few specific messages (for example, "Configuration Approved" and "System Update Successful"). After seeing the public key, they find a way to forge valid signatures only for those specific messages.
- b) During a penetration testing session, a student manages to create one valid message-signature pair using only Riya's public key. However, the student has no control over which message the forged signature corresponds to-it is valid for some arbitrary message.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) Passwords remain one of the most widely used forms of authentication despite their weaknesses. (5)
- (a) Describe the key characteristics of a strong password and outline at least three common methods attackers use to compromise passwords.
- (b) A large enterprise has just been breached because employees reused weak passwords across multiple services. Recommend a comprehensive password policy for the enterprise, detailing at least four components (such as length, complexity, rotation, reuse prevention, storage practices), and justify how each component mitigates risk.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) Explain the working of the TLS Record Protocol and its role in providing both confidentiality and message integrity. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4C) Compare symmetric-key challenge-response authentication with asymmetric challenge-response methods. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) During an online banking transaction, sensitive user credentials are transmitted from the client to the bank's web server. Explain how the SSL Record Protocol in this situation ensures confidentiality and integrity of the transmitted data. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- 5B) A company firewall supports inbound HTTP and SSH but blocks ICMP. Evaluate its security posture and suggest additional IDS configuration improvements. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5C) Using the Paillier cryptosystem with $(n = 33, g = 4)$, decrypt ciphertext $c = 25$. Show step-by-step derivation. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 IT_3121:INFORMATION SECURITY

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	7
3	3	CO 3	8
4	4	CO 4	22
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 IT_3121:INFORMATION SECURITY

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	18(36%)
4	Analysis	7	27(54%)
5	Create	1	5(10%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)